PROMPT ENGINEERING LAB EXPERIMENT

Aim: To test and compare how different pattern models respond to various prompts (broad or unstructured) versus basic prompts (clearer and more refined) across multiple scenarios. Analyze the quality, accuracy, and depth of the generated responses

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1. Education Scenario

A college student is preparing for an examination on blockchain technology. The student uses an AI tool to understand complex topics in simple language. The AI explains the concept, gives examples, and provides a short summary so the student can revise quickly and improve understanding before the exam.

Basic response:

Example 1
Question: Explain the Internet in simple words.
Answer: The Internet is a global network that connects computers and devices around the world. It allows people to share information, communicate, and access websites, emails, and online services quickly.

Example 2
Question: Explain Cloud Computing in simple words.
Answer: Cloud computing is a technology that lets people store data and run applications on remote servers through the internet instead of on their personal computers.

Actual Question: Explain Blockchain Technology in simple words.

Basic Response:
Blockchain is a digital system used to store information securely. Data is stored in blocks that are linked together in a chain. Each block contains records that cannot be easily changed. Many computers share the same data, making the system transparent and secure. It is widely used in cryptocurrencies like Bitcoin.

Enhanced response:

Example 1
Input: Explain Artificial Intelligence in simple terms with an example and a short summary.
Output: Artificial Intelligence (AI) is technology that allows machines to learn from data and perform tasks that usually require human intelligence. For example, voice assistants like Siri or Alexa answer questions and perform tasks.
Summary: AI helps machines think and learn to solve problems.

Example 2
Input: Explain Cloud Computing in simple terms with an example and a short summary.
Output: Cloud computing is a technology that allows users to store data and run applications over the internet instead of using a local computer. For example, Google Drive lets people store and access files online.
Summary: Cloud computing helps users access data anytime from anywhere.

Actual Prompt
Input: Explain Blockchain Technology in simple terms with an example and a short summary.

Enhanced Response:
Blockchain is a digital system used to record and store information securely across many computers. Instead of keeping data in one place, it is shared across a network, making it difficult to change or hack. Each block stores data and is connected to the previous block, forming a chain.

Example: Cryptocurrency transactions like Bitcoin are recorded using blockchain technology.

Summary: Blockchain is a secure and transparent way to store and share digital information.

Zero-Shot Prompting (Basic Response)

Prompt: Explain blockchain technology in simple language with an example and a short summary.

Response:
Blockchain technology is a digital system used to store and record information securely. Instead of keeping data in one central place, the information is stored across many computers called nodes. The data is organized into blocks, and these blocks are linked together to form a chain. Once information is recorded, it is very difficult to change, which makes blockchain secure and transparent.

Example:
Bitcoin transactions are recorded on a blockchain, where every transaction is verified and stored in blocks.

Summary:
Blockchain is a decentralized and secure technology used to store and share digital information.

Zero-Shot Prompting (Enhanced Response)

Prompt: Explain blockchain technology in simple language. Include an example and a short summary.

Enhanced Response:
Blockchain technology is a digital system used to record and store information securely across a network of computers. Instead of storing data in a single central database, blockchain distributes the data among many computers called nodes. The information is stored in blocks, and each block is connected to the previous one,

forming a chain. Once a block is added, it cannot be easily changed, which ensures security, transparency, and trust in the system.

Example:
When a person sends Bitcoin to another person, the transaction is verified by the network and recorded in a block. After verification, the block is added to the blockchain, making the transaction permanent and visible to all participants in the network.

Summary:
Blockchain is a decentralized and secure method of recording digital transactions. It improves transparency, reduces the risk of data tampering, and is widely used in cryptocurrencies, finance, and other digital applications.

Chain-of-Thought Prompting (Basic Response)

Prompt: Explain blockchain technology in simple language with reasoning, an example, and a short summary.

Response (Step-by-Step Explanation):

Step 1 – What is Blockchain?
Blockchain is a digital technology used to store and record information securely. The data is stored in blocks, and these blocks are connected together in a chain.

Step 2 – How it Works
When new information or a transaction is created, it is placed into a block. This block is then verified by a network of computers and added to the chain. Once added, the information cannot be easily changed.

Step 3 – Example
For example, when someone sends Bitcoin to another person, the transaction is verified and stored in a block on the blockchain.

Step 4 – Summary
Blockchain is a secure and decentralized system used to record and share digital information. It is commonly used in cryptocurrencies and other digital applications.

Chain-of-Thought Prompting (Enhanced Response)

Prompt: Explain blockchain technology in simple language. Provide step-by-step reasoning, an example, and a short summary.

Response (Step-by-Step Explanation):

Step 1 – Understanding the Concept
Blockchain technology is a digital system used to record and store information securely. Instead of storing data in one central place, the information is shared across many computers in a network.

Step 2 – Structure of Blockchain
The information is stored in units called **blocks**. Each block contains data, a timestamp, and a link to the previous block. When blocks are connected together, they form a **chain**, which is why it is called blockchain.

Step 3 – How the System Works
When a new transaction or piece of information is created, it is first verified by multiple computers in the network, called nodes. After verification, the transaction is added to a new block and connected to the existing chain. Once recorded, the data cannot be easily changed.

Step 4 – Security and Transparency
Because the data is distributed across many computers and each block is linked to the previous one, it is very difficult to alter the information. This makes blockchain secure, transparent, and trustworthy.

Step 5 – Example
For example, when a person sends **Bitcoin** to another person, the transaction is verified by the network and recorded on the blockchain. Everyone in the network can see the transaction, but they cannot change it.

Summary:
Blockchain is a decentralized technology used to store and share digital information securely. It records data in connected blocks, making it transparent, reliable, and widely used in cryptocurrencies and other digital systems.

Scenario 2:

Programming Scenario A beginner programmer is learning the C programming language and asks an AI tool to explain pointers and recursion with simple code examples to help understand the concepts.

Few-Shot Prompting (Basic Response)

Example 1
Input: Explain arrays in C with a simple example.
Output: An array in C is a collection of elements of the same data type stored in contiguous memory locations. It allows programmers to store multiple values in a single variable name.
Example:

```
int numbers[3] = {10, 20, 30};
printf("%d", numbers[0]);
```

Example 2
Input: Explain functions in C with a simple example.
Output: A function in C is a block of code that performs a specific task. It helps organize programs and reuse code.
Example:

```
int add(int a, int b) {
    return a + b;
}
```

Actual Prompt
Input: Explain pointers and recursion in C with simple examples.

Basic Response:

Pointers:
A pointer is a variable that stores the memory address of another variable.

Example:

```
int x = 10;
int *p;
p = &x;
printf("%d", *p);
```

Recursion:
Recursion is when a function calls itself to solve a problem.

Example:

```
int factorial(int n) {
    if(n == 0)
        return 1;
    else
        return n * factorial(n - 1);
}
```

Few-Shot Prompting (Enhanced Response)

Example 1
Input: Explain arrays in C with a simple example and short explanation.
Output:
An array in C is a collection of elements of the same data type stored in contiguous memory locations. Arrays help store multiple values using a single variable name.

Example:

```
#include <stdio.h>
int main() {
    int numbers[3] = {10, 20, 30};
    printf("%d", numbers[1]);
    return 0;
}
```

Explanation: The array stores three integers, and numbers[1] accesses the second element.

Example 2
Input: Explain functions in C with a simple example and short explanation.
Output:
A function in C is a block of code that performs a specific task. Functions help make programs organized and reusable.

Example:

```
#include <stdio.h>
int add(int a, int b) {
    return a + b;
}
int main() {
    printf("%d", add(5, 3));
    return 0;
}
```

Explanation: The function add() takes two numbers and returns their sum.

Actual Prompt
Input: Explain pointers and recursion in C with simple examples.

Enhanced Response:

Pointers:
A pointer is a variable that stores the memory address of another variable. It helps in accessing and modifying data stored in memory.

Example:

```
#include <stdio.h>
int main() {
    int x = 10;
    int *p = &x;

    printf("Value of x: %d\n", x);
    printf("Address of x: %p\n", &x);
    printf("Value using pointer: %d\n", *p);

    return 0;
}
```

Explanation:
The pointer p stores the address of variable x. Using *p, we can access the value stored in x.

Recursion:
Recursion is a technique where a function calls itself to solve a problem step by step until a base condition is reached.

Example (Factorial):

```
#include <stdio.h>
int factorial(int n) {
    if(n == 0)
        return 1;
    else
        return n * factorial(n - 1);
}

int main() {
    printf("Factorial: %d", factorial(5));
    return 0;
}
```

Explanation:
The function repeatedly calls itself with a smaller value until it reaches the base condition (n == 0).

Zero-Shot Prompting (Basic Response)

Prompt: Explain pointers and recursion in C programming with simple code examples.

Response:

Pointers:
A pointer in C is a variable that stores the memory address of another variable. Pointers help in accessing and modifying data directly in memory.

Example:

```
#include <stdio.h>
int main() {
    int x = 10;
    int *p = &x;

    printf("Value of x: %d\n", x);
    printf("Value using pointer: %d\n", *p);

    return 0;
}
```

Here, p stores the address of x, and *p is used to access the value of x.

Recursion:
Recursion is a programming technique where a function calls itself to solve a problem until a base condition is reached.

Example:

```
#include <stdio.h>
int factorial(int n) {
    if (n == 0)
        return 1;
    else
        return n * factorial(n - 1);
}

int main() {
    printf("Factorial of 5 = %d", factorial(5));
    return 0;
}
```

Zero-Shot Prompting (Enhanced Response)

Prompt: Explain pointers and recursion in C programming with simple code examples and a short explanation.

Pointers

A **pointer** in C is a variable that stores the **memory address** of another variable. Instead of directly holding a value, it points to the location where the value is stored in memory. Pointers are useful for memory management, passing arguments to functions, and working with arrays and data structures.

Example:

```
#include <stdio.h>

int main() {
    int x = 10;       // normal variable
    int *p = &x;      // pointer storing the address of x

    printf("Value of x: %d\n", x);
    printf("Address of x: %p\n", &x);
    printf("Value using pointer: %d\n", *p);

    return 0;
}
```

Explanation:

- int *p declares a pointer variable.
- &x gives the **address** of variable x.
- *p is used to **access the value stored at that address**.

Recursion

Recursion is a programming technique where a **function calls itself** to solve a problem. It breaks a large problem into smaller sub-problems. Every recursive function must have a **base condition** to stop the repeated calls.

Example (Factorial Calculation):

```
#include <stdio.h>

int factorial(int n) {
    if (n == 0 || n == 1)   // base condition
        return 1;
    else
        return n * factorial(n - 1); // recursive call
}
```

```
int main() {
    int num = 5;
    printf("Factorial of %d = %d", num, factorial(num));
    return 0;
}
```

Explanation:

- The function calls itself with n-1.
- The process continues until the base condition (n == 0 or 1) is reached.
- Then the results are returned step by step.

Summary

- **Pointers** store memory addresses and allow direct access to data in memory.
- **Recursion** is a technique where a function repeatedly calls itself to solve a problem until a stopping condition is reached.

Both concepts are important in **C programming** for efficient memory handling and solving complex problems.

Chain-of-Thought Prompting (Basic Response)

Prompt: Explain pointers and recursion in C programming with step-by-step reasoning and simple code examples.

Response:

Step 1 – Understanding Pointers
A pointer is a variable that stores the memory address of another variable. It allows a programmer to directly access and modify the value stored in memory.

Example:

```
#include <stdio.h>
int main() {
    int x = 10;
    int *p = &x;

    printf("Value of x = %d\n", x);
    printf("Value using pointer = %d\n", *p);

    return 0;
}
```

Here, p stores the address of x, and *p is used to access the value stored at that address.

Step 2 – Understanding Recursion
Recursion is a programming technique where a function calls itself repeatedly to solve a problem until a stopping condition (base case) is reached.

Example:

```
#include <stdio.h>
int factorial(int n) {
   if (n == 0)
      return 1;
   else
      return n * factorial(n - 1);
}

int main() {
   printf("Factorial = %d", factorial(5));
   return 0;
}
```

Step 3 – Summary
Pointers help store and access memory addresses, while recursion allows functions to repeatedly call themselves to solve problems step by step.

Chain-of-Thought Prompting (Enhanced Response)

Prompt: Explain pointers and recursion in C programming with step-by-step reasoning, simple code examples, and a short summary.

Step 1 – Understanding Pointers

A **pointer** is a special variable in C that stores the **memory address** of another variable. Instead of directly storing a value, the pointer refers to the location where the value is stored. This helps programmers work with memory efficiently and modify values indirectly.

Step 2 – How Pointers Work

1. A normal variable stores a value.
2. A pointer variable stores the **address** of that variable.

3. Using the **dereference operator (*)**, we can access the value stored at that memory location.

Example:

```
#include <stdio.h>

int main() {
    int x = 20;      // normal variable
    int *p = &x;     // pointer storing the address of x

    printf("Value of x: %d\n", x);
    printf("Address of x: %p\n", &x);
    printf("Value using pointer: %d\n", *p);

    return 0;
}
```

Explanation:

- &x gives the memory address of x.
- int *p declares a pointer variable.
- *p accesses the value stored at that address.

Step 3 – Understanding Recursion

Recursion is a programming technique where a function **calls itself** to solve a problem. It breaks a large problem into smaller parts until a **base condition** stops the repeated calls.

Step 4 – How Recursion Works

1. A function calls itself with a smaller input.
2. This continues until the **base condition** is reached.
3. Then the function returns values step by step.

Example (Factorial Program):

```
#include <stdio.h>

int factorial(int n) {
    if (n == 0 || n == 1)   // base condition
        return 1;
```

```
    else
        return n * factorial(n - 1); // recursive call
}

int main() {
    int num = 5;
    printf("Factorial of %d = %d", num, factorial(num));
    return 0;
}
```

Explanation:
The function keeps calling itself with n-1 until n becomes 1, then it multiplies the values while returning.

Step 5 – Summary

- **Pointers** store memory addresses and allow direct access to data stored in memory.
- **Recursion** is a method where a function repeatedly calls itself until a stopping condition is reached.
- Both concepts are essential in **C programming** for efficient memory management and solving complex problems.

Scenario 3:

Business Scenario A small business owner wants to promote their shop on social media. They use an AI tool to generate a short marketing caption and hashtags for a new product launch. for this scenario generate a basic response in few shot prompt

Few-Shot Prompting (Basic Response)

Example 1
Input: Create a short social media caption and hashtags for a new coffee shop.
Output:
Caption: Enjoy the rich aroma of freshly brewed coffee at our new coffee shop. Visit us today for a refreshing experience!
Hashtags: #CoffeeLovers #FreshBrew #CoffeeTime #CafeVibes

Example 2
Input: Create a short social media caption and hashtags for a clothing store sale.
Output:
Caption: Upgrade your style with our latest fashion collection. Shop now and enjoy amazing discounts!
Hashtags: #FashionSale #NewCollection #StyleTrend #ShopNow

Actual Prompt
Input: Create a short marketing caption and hashtags for a new product launch in a small shop.

Basic Response:
Caption: Exciting news! Our new product has just arrived. Come and check it out today and experience something amazing.

Few-Shot Prompting (Enhanced Response)

Example 1
Input: Create a short social media caption and hashtags for a new coffee shop opening.
Output:
Caption: Start your day with the perfect cup of coffee! Visit our new coffee shop and enjoy freshly brewed flavours in a cozy atmosphere.
Hashtags: #CoffeeTime #FreshBrew #CafeVibes #CoffeeLovers #MorningEnergy

---

Example 2
Input: Create a short social media caption and hashtags for a clothing store launching a new collection.
Output:
Caption: Step into style with our brand-new fashion collection. Trendy designs, great quality, and amazing offers waiting just for you!
Hashtags: #NewCollection #FashionStyle #ShopNow #TrendAlert #StyleInspiration

---

Actual Prompt
Input: Create a short marketing caption and hashtags for a new product launch in a small shop.

Enhanced Response:
Caption:
We’re excited to introduce our **latest product**! Discover quality, style, and value all in one place. Visit our shop today and be among the first to try our newest arrival. Don’t miss out!

Hashtags:
#NewLaunch #NewArrival #ShopLocal #SmallBusiness #ProductLaunch #SupportLocal #TrendingNow #MustTryProduct

Short Note:
This caption attracts attention, highlights the new product, and encourages customers to visit the shop while hashtags help increase visibility on social media.

Zero-Shot Prompting (Basic Response)

Prompt: Generate a short social media marketing caption and hashtags for a small business launching a new product.

Response:
Caption:
We are excited to announce the launch of our brand-new product! Visit our shop today and discover something special made just for you. Don't miss the chance to try our latest arrival.

Hashtags:
#NewProduct #NewLaunch #ShopLocal #SmallBusiness #NewArrival #SupportLocal

Zero-Shot Prompting (Enhanced Response)

Prompt: Generate a short social media marketing caption and hashtags for a small business launching a new product.

Enhanced Response:

Caption:
We're excited to introduce our **brand-new product**! Designed with quality and style in mind, it's the perfect addition to our collection. Visit our shop today and be one of the first to experience it. Hurry in and discover something new and exciting!

Hashtags:
#NewProductLaunch #NewArrival #ShopLocal #SmallBusiness #SupportLocalBusiness #TrendingNow #MustTry #CustomerFavorites

Short Explanation:
The caption highlights the excitement of the product launch and encourages customers to visit the shop, while the hashtags help increase reach and visibility on social media platforms.

Chain-of-Thought Prompting (Basic Response)

Prompt: Generate a short social media marketing caption and hashtags for a new product launch for a small business.

Response (Step-by-Step):

Step 1 – Identify the goal
The goal is to promote a new product and attract customers to the shop through social media.

Step 2 – Create an engaging caption
The caption should be short, attractive, and encourage people to check out the new product.

Caption:
We’re excited to launch our brand-new product! Visit our shop today and discover something special made just for you.

Step 3 – Add relevant hashtags
Hashtag help reach more people and increase visibility on social media.

Hashtags:
#NewProduct #ProductLaunch #ShopLocal #SmallBusiness #NewArrival

Step 4 – Summary
A short caption and relevant hashtags help promote the new product and attract more customers online. 🚀

Chain-of-Thought Prompting (Enhanced Response)

Prompt: Generate a short social media marketing caption and hashtags for a new product launch for a small business.

Step 1 – Understand the Goal

The main goal is to **promote a new product** and **attract customers** through social media. The message should be simple, engaging, and encourage people to visit the shop or try the new product.

Step 2 – Identify Key Marketing Elements

To create an effective promotion, the caption should include:

- Excitement about the **new product launch**
- A **clear call to action** (visit the shop or check it out)
- Positive and attractive wording to grab attention.

Step 3 – Create an Engaging Caption

Caption:
🎉 Big news! Our **brand-new product** has just arrived. Be among the first to

explore it and experience something fresh and exciting. Visit our shop today and discover your new favourite item!

Step 4 – Add Effective Hashtags

Hashtags help the post reach more people and increase engagement.

Hashtags:
#NewProductLaunch #NewArrival #ShopLocal #SmallBusiness #SupportLocal #TrendingNow #ProductLaunch #MustTry

---

Step 5 – Summary

A clear caption combined with relevant hashtags helps promote the new product effectively, increases online visibility, and encourages more customers to visit the shop. 🚀

Scenario 4:

A **patient receives a medical report** but does not understand the technical terms. They ask an AI assistant to **explain the report in simple words** and tell whether the results look normal.

Few-Shot Prompting (Basic Response)

Example 1
Input: Explain a blood pressure report in simple words.
Output: Blood pressure shows how strongly blood pushes against the walls of the arteries. A normal reading is around **120/80 mmHg**. If the value is close to this range, it usually means the blood pressure is normal.

Example 2
Input: Explain a blood sugar test result in simple words.
Output: A blood sugar test measures the amount of glucose in the blood. A fasting blood sugar level between **70–100 mg/dL** is usually considered normal. Higher levels may indicate high blood sugar.

Actual Prompt
Input: Explain a medical report in simple words and tell whether the results look normal.

Basic Response:
A medical report contains the results of different health tests. Each result is compared with a **normal reference range**. If the value is within the range, it usually

means the result is normal. If it is higher or lower than the range, it may need medical attention. It is always best to consult a doctor for proper advice.

Few-Shot Prompting (Enhanced Response)

Example 1
Input: Explain a blood pressure report in simple words and say if it is normal.
Output:
Blood pressure measures how strongly blood flows through your arteries. A normal blood pressure level is around **120/80 mmHg**. If your reading is close to this range, it usually means your blood pressure is normal. If it is much higher or lower, it may require medical attention.

Example 2
Input: Explain a blood sugar test report in simple words and say if it is normal.
Output:
A blood sugar test checks the amount of glucose in the blood. A **fasting blood sugar level between 70–100 mg/dL** is generally considered normal. If the value is higher, it may indicate high blood sugar, and if it is lower, it may indicate low blood sugar.

Actual Prompt
Input: Explain a medical report in simple words and tell whether the results look normal.

Enhanced Response:
A medical report contains the results of different health tests such as blood count, sugar level, cholesterol, or other measurements. Each result is compared with a **reference range**, which shows the normal values for healthy people.

If the test results are **within the reference range**, it usually means the results are normal and the body is functioning properly. If some values are **higher or lower than the normal range**, it may indicate a possible health issue that needs attention.

For example, if **hemoglobin levels** are within the normal range, it means the blood is carrying oxygen properly. If it is below the normal range, it may indicate anemia.

Summary:
Most results within the reference range suggest normal health, but a doctor should always review the report to provide the correct medical advice and diagnosis.

Zero-Shot Prompting (Basic Response)

Prompt: Explain a medical report in simple words and tell whether the results look normal.

Response:
A medical report shows the results of different health tests. Each test result is compared with a **normal reference range**, which shows the values expected in healthy people.

If the result is **within the normal range**, it usually means the result is normal. If the value is **higher or lower than the range**, it may indicate a health problem and further medical advice may be needed.

For example, if your **hemoglobin level** is within the normal range, it means your blood is functioning properly.

Summary:
If most values are within the reference range, the results generally look normal. However, it is always best to consult a doctor for a proper medical interpretation.

Zero-Shot Prompting (Enhanced Response)

Prompt: Explain a medical report in simple words and tell whether the results look normal.

Enhanced Response:
A medical report contains the results of different health tests such as blood count, sugar level, cholesterol, and other body measurements. These results help doctors understand how well your body is functioning.

Each test result is compared with a **reference range**, which represents the normal values for healthy individuals. If your test result is **within this range**, it usually means that part of your health is normal. If a result is **above or below the normal range**, it may indicate a possible health issue that requires further attention.

For example, if your **hemoglobin level** is within the normal range, it means your blood can carry oxygen properly. If the level is lower than normal, it may suggest anemia. Similarly, normal blood sugar levels indicate that your body is managing glucose properly.

Summary:
If most values in the report fall within the reference ranges, the results generally look normal. However, only a qualified doctor can give an accurate diagnosis and medical advice after reviewing the complete report and medical history.

Chain-of-Thought Prompting (Basic Response)

Prompt: Explain a medical report in simple words and indicate whether the results appear normal.

Response (Step-by-Step Explanation):

Step 1 – Understanding the Medical Report
A medical report contains the results of different health tests such as blood tests, sugar levels, and cholesterol levels.

Step 2 – Compare with Normal Range
Each result is compared with a **reference range**, which shows the normal values for healthy people.

Step 3 – Identify Normal or Abnormal Results
If the test values are **within the normal range**, they are usually considered normal. If they are **higher or lower**, it may indicate a health issue.

Step 4 – Example
For example, if the **hemoglobin level** is within the normal range, it means the blood is working properly.

Step 5 – Summary
Most values within the normal range suggest the results look normal, but a **doctor should review the report** for accurate medical advice.

Chain-of-Thought Prompting (Enhanced Response)

Prompt: Explain a medical report in simple words and indicate whether the results appear normal.

Step 1 – Understanding the Medical Report

A medical report contains the results of different health tests such as blood count, blood sugar, cholesterol, and other body measurements. These tests help doctors understand how well different parts of the body are functioning.

Step 2 – Checking the Reference Range

Each result in the report is compared with a **reference range**, which shows the normal values for healthy people. These ranges help identify whether the test results are normal or need attention.

Step 3 – Interpreting the Results

If the test values are **within the reference range**, it usually means that the result is normal and the body is functioning properly. If a value is **higher or lower than the range**, it may indicate a possible health issue that may require further testing or treatment.

Step 4 – Example

For example, if the **hemoglobin level** in the report is within the normal range, it means the blood is carrying oxygen properly. If the value is lower than normal, it may suggest anemia. Similarly, normal blood sugar levels indicate good glucose control in the body.

Step 5 – Summary

Most results falling within the reference ranges suggest that the report looks normal. However, a **doctor should always review the full report** along with the patient's medical history to provide an accurate diagnosis and proper medical advice.

Scenario 5:

A **student preparing a presentation** asks an AI tool to **create PPT slide content about roofing technology with headings and bullet points**.

Few-Shot Prompting (Basic Response)

Example 1
Input: Create PPT slide content about Artificial Intelligence with headings and bullet points.
Output:
Slide Title: Artificial Intelligence

- AI is a technology that enables machines to simulate human intelligence.
- It helps computers learn from data and make decisions.
- AI is used in voice assistants, robotics, and automation.

Example 2
Input: Create PPT slide content about Cloud Computing with headings and bullet points.
Output:
Slide Title: Cloud Computing

- Cloud computing allows users to store and access data over the internet.
- It provides services like storage, software, and servers online.
- Examples include Google Drive and Microsoft OneDrive.

Actual Prompt
Input: Create PPT slide content about roofing technology with headings and bullet points.

Basic Response:

Slide Title: Roofing Technology

- Roofing technology refers to modern methods used to design and construct roofs.
- It helps protect buildings from weather conditions such as rain, heat, and wind.
- Different materials like metal, tiles, and shingles are used in roofing.
- Modern roofing focuses on durability, energy efficiency, and sustainability.

Few-Shot Prompting (Enhanced Response)

Example 1
Input: Create PPT slide content about Artificial Intelligence with headings and bullet points.

Output:
Slide Title: Artificial Intelligence

- Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence.
- AI systems learn from data and improve their performance over time.
- Applications include virtual assistants, robotics, and smart recommendation systems.
- AI is widely used in healthcare, finance, and transportation.

Example 2
Input: Create PPT slide content about Cloud Computing with headings and bullet points.

Output:
Slide Title: Cloud Computing

- Cloud computing provides storage, software, and services through the internet.
- Users can access data and applications from anywhere.
- It reduces the need for physical hardware and maintenance.

- Examples include Google Drive, Microsoft OneDrive, and Amazon Web Services.

Actual Prompt
Input: Create PPT slide content about roofing technology with headings and bullet points.

Enhanced Response:

Slide Title: Roofing Technology

Introduction

- Roofing technology refers to modern techniques and materials used in roof construction.
- It plays an important role in protecting buildings from environmental conditions.

Types of Roofing Materials

- Metal roofing
- Clay or concrete tiles
- Asphalt shingles
- Green roofing systems

Advantages of Modern Roofing

- Provides better durability and long-lasting protection.
- Improves energy efficiency in buildings.
- Helps reduce maintenance costs.

Applications

- Used in residential, commercial, and industrial buildings.
- Supports sustainable and eco-friendly construction practices.

Zero-Shot Prompting (Basic Response)

Prompt: Create PPT slide content about roofing technology with headings and bullet points.

Response:

Slide 1: Introduction to Roofing Technology

- Roofing technology refers to modern methods used in building and designing roofs.
- It helps protect buildings from rain, heat, wind, and other weather conditions.

Slide 2: Types of Roofing Materials

- Metal roofing
- Clay and concrete tiles
- Asphalt shingles
- Slate roofing

Slide 3: Benefits of Modern Roofing

- Provides strong protection for buildings
- Improves durability and lifespan of roofs
- Helps in energy efficiency

Slide 4: Applications of Roofing Technology

- Residential buildings
- Commercial buildings
- Industrial structures

Slide 5: Conclusion

- Roofing technology improves building safety and durability.
- Modern roofing materials help create sustainable and efficient structures.

Zero-Shot Prompting (Enhanced Response)

Prompt: Create PPT slide content about roofing technology with headings and bullet points.

Slide 1: Introduction to Roofing Technology

- Roofing technology involves modern methods and materials used to construct and maintain roofs.
- It protects buildings from environmental factors like rain, sunlight, wind, and temperature changes.
- Modern roofing systems focus on durability, safety, and energy efficiency.

Slide 2: Types of Roofing Materials

- **Metal Roofing:** Strong, durable, and long-lasting.
- **Clay and Concrete Tiles:** Commonly used for traditional and durable roofing.
- **Asphalt Shingles:** Affordable and widely used in residential buildings.
- **Green Roofs:** Eco-friendly roofs covered with vegetation.

Slide 3: Advantages of Modern Roofing

- Improves the strength and durability of buildings.
- Provides better insulation and energy efficiency.
- Reduces maintenance costs over time.
- Enhances the appearance and value of buildings.

Slide 4: Applications of Roofing Technology

- Used in **residential homes** for safety and protection.
- Applied in **commercial buildings** such as offices and malls.
- Used in **industrial structures** like factories and warehouses.

Slide 5: Conclusion

- Roofing technology plays a vital role in modern construction.
- Advanced materials and techniques improve safety, sustainability, and energy efficiency.
- Choosing the right roofing system increases the lifespan and performance of buildings.

Chain-of-Thought Prompting (Basic Response)

Prompt: Create PPT slide content about roofing technology with headings and bullet points.

Response (Step-by-Step):

Step 1 – Identify the Topic
The topic is **roofing technology**, which includes methods and materials used in constructing roofs to protect buildings.

Step 2 – Create an Introduction Slide
Slide 1: Introduction to Roofing Technology

- Roofing technology refers to modern techniques used in roof construction.
- It protects buildings from weather conditions like rain, sun, and wind.

Step 3 – Explain Types of Roofing Materials
Slide 2: Types of Roofing Materials

- Metal roofing
- Clay tiles
- Asphalt shingles
- Concrete roofing

Step 4 – Mention Benefits of Roofing Technology
Slide 3: Advantages of Roofing Technology

- Provides protection and durability
- Improves energy efficiency
- Increases the lifespan of buildings

Step 5 – Conclusion Slide
Slide 4: Conclusion

- Roofing technology is important in modern construction.
- It helps create strong, safe, and long-lasting buildings.

Chain-of-Thought Prompting (Enhanced Response)

Prompt: Create PPT slide content about roofing technology with headings and bullet points.

Step 1 – Understand the Topic

The topic **roofing technology** refers to the materials, designs, and construction techniques used to build strong and durable roofs that protect buildings from weather and environmental conditions.

Step 2 – Create the Introduction Slide

Slide 1: Introduction to Roofing Technology

- Roofing technology involves modern techniques used in roof construction and design.
- It protects buildings from rain, sunlight, wind, and extreme temperatures.
- Modern roofing focuses on durability, safety, and energy efficiency.

Step 3 – Types of Roofing Materials

Slide 2: Types of Roofing Materials

- **Metal Roofing:** Durable, lightweight, and long-lasting.
- **Clay or Concrete Tiles:** Strong and commonly used in residential buildings.
- **Asphalt Shingles:** Affordable and widely used roofing material.
- **Green Roofs:** Eco-friendly roofs covered with plants and vegetation.

Step 4 – Advantages of Roofing Technology

Slide 3: Advantages of Modern Roofing

- Improves building durability and protection.
- Enhances energy efficiency by reducing heat absorption.
- Requires less maintenance and increases roof lifespan.
- Improves the appearance and value of buildings.

Step 5 – Applications and Conclusion

Slide 4: Applications of Roofing Technology

- Residential homes
- Commercial buildings such as offices and malls
- Industrial buildings like factories and warehouses

Slide 5: Conclusion

- Roofing technology plays an important role in modern construction.
- Advanced materials and techniques help build safer, stronger, and more sustainable structures.

SCENARIO	BASIC RESPONSE			ENHANCED RESPONSE		
	Evaluation	Few shot		Zero shot		Chain of thought
1	Accuracy:	4	5	3	4	5	3
	speed	3	4.5	4	5	3	3.5
	Coherence	4	5	4	3.5	4	5
	UX	3	4	3	4	3	4.5
2	Accuracy	4	5	3	3.5	5	4
	speed	3	4.5	4	3.5	3	3.5
	coherence	5	5	4.5	3	4	3
	UX	4.5	3.5	4	4	5	3.5
3	Accuracy	4	4.5	4	5	5	4
	speed	3	4	5	3	4.5	3
	coherence	5	3	4	4	3.5	4
	UX	4.5	3	5	4	4	4.5
4	Accuracy	5	4	4.5	5	4.5	5
	speed	3	4	4.5	3	3.5	3
	coherence	3.5	5	3	3.5	5	4
	UX	4.5	4.5	3.5	4	4	3.5
5	Accuracy	4	5	4.5	4.5	4	5
	speed	3.5	4.5	5	5	3.5	4.5
	Coherence	5	4	3.5	4.5	4	5
	UX	4.5	3.5	4	5	5	4.5